

MANSI GANESH MENGDE

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OBJECTIVE

Data Engineer specializing in **cloud data platform architecture**, **ETL/ELT pipeline modernization** and **enterprise data warehousing** with shipping production data systems in regulated, high throughput environments at scale. Owned **data platform migrations**, **governance frameworks** and **multi source data ingestion** across nuclear infrastructure, enterprise network security and institutional analytics using **SQL**, **PySpark**, **Databricks**, **Kafka**, **Airflow** and **dbt**, driven to replace vendor dependent data solutions with owned, audit ready, standardized platforms.

EDUCATION

California State University, Long Beach	Sep 2024 – May 2026
M.S. Information Systems	
International Institute of Information Technology, Pune	Dec 2021 – Jul 2024
B.E. Information Technology	

WORK EXPERIENCE

Research Assistant, AI Systems and Agentic Workflows	Sep 2024 – May 2026
Institutional Research and Analytics (IR&A), California State University Long Beach	Long Beach, CA
- Owned full data platform architecture for institutional analytics, designing BI oriented schemas, ETL pipelines and warehouse structures in Python, SQL and PostgreSQL, standardizing raw admissions, financial aid and academic records pulled from Excel exports, PDF reports and 5 upstream source systems into trusted, audit ready datasets for 200 non technical stakeholders, cutting analyst turnaround from 5 days to same day. - Established data governance, integrity and standardization framework tracking schema drift, pipeline freshness and data quality across 5 source systems, implemented GitHub Actions CI/CD with automated test gates as deployment blockers before data reached production, and drove tool selection decisions that raised platform reliability to 99.9% accuracy across 3 reporting cycles. - Built end to end data ingestion and structuring pipeline converting unstructured PDF records and raw text into queryable, governed datasets using automated extraction and validation, achieving 99.2% F1 score on PII redaction and improving downstream data accuracy from 61% to 89% through structured testing across 3 staged production releases.	
Software Development Engineer I	Nov 2023 – May 2024
AtkinsRealis (Global Engineering and Nuclear Infrastructure, \$9.7B Revenue)	Maharashtra, IN
Core Team: Mission Critical Asset Automation Group - Architected and owned scalable data platform and ETL pipeline orchestration using PySpark on Databricks and AWS S3, ingesting 5 million sensor readings per day from 20 source systems, driving schema design, data modeling and partition tuning decisions that cut pipeline runtime by 40% and sustained zero data outages over 6 months across nuclear and energy infrastructure reporting. - Modernized data observability and quality standards for asset pipelines using Prometheus and Grafana, implemented anomaly detection via Isolation Forest over PySpark cutting false alerts by 28%, and provisioned pipeline infrastructure as code via Terraform and Jenkins CI/CD, enforcing governance gates across distributed engineering teams transitioning legacy systems to owned infrastructure. - Optimized SQL query performance and relational data access layers via Hibernate, JPA and advanced indexing for enterprise asset management, cutting API latency by 24%, and built data structuring pipeline for unstructured logs that ingested raw text into standardized formats, cutting reporting time from 48 hours to under 1 hour.	
Data Engineer	Dec 2022 – Aug 2023
AlgoSec (AI Powered Network Security Policy Management)	Maharashtra, IN
Core Team: Enterprise Scale Security Analytics and Ingestion Lab - Designed and built scalable data warehouse and ELT pipeline architecture using Python, dbt and Airflow across 500+ enterprise customer environments, structuring raw multi format source data into governed, BI oriented PostgreSQL schemas that powered self serve analytics and compliance reporting at 99.9% pipeline uptime. - Established data quality and integrity framework using Great Expectations as automated blocking gates on every ingested record, instrumented pipeline telemetry via Prometheus to catch data regressions within 2 minutes, and processed 1.2 million daily events through Kafka streaming cutting reporting latency from 48 hours to under 1 hour. - Partnered directly with business stakeholders and technical leadership to define data architecture strategy for behavioral analytics on 1.2 million daily records, driving tool and platform decisions across feature engineering and pipeline design that surfaced business critical signals 3 weeks earlier, supporting 15% retention improvement across enterprise SaaS contracts.	

PROJECTS

AutonomousOps: Self Healing Multi Agent Infrastructure Operations Platform [GitHub] [Live]
Python, Kafka, pgvector, FastAPI, Kubernetes, Prometheus, Grafana, Slack API
Designed and owned end to end data platform ingesting live telemetry via Kafka streaming, structuring raw operational signals into a governed, queryable data layer over pgvector with built in data observability, lineage and quality monitoring, replacing manual triage with an owned, automated pipeline that cut mean time to resolution by 74% and prevented 91% of repeat incidents across 200 production scenarios.
VisionRAG: Multimodal Document Intelligence and Data Ingestion Pipeline [GitHub] [Live]
Python, pgvector, FastAPI, AWS S3, Docker, React
Built full stack data ingestion and structuring pipeline converting 50K+ raw PDF and image documents per day into standardized, governed datasets with end to end data quality checks, lineage tracking and observability, achieving 94% extraction accuracy vs 76% baseline, deployed on AWS S3 and Docker, cutting manual document processing by 85%.

TECHNICAL SKILLS

Languages and Frameworks: Python, TypeScript, FastAPI, REST APIs, asyncio, Java Spring Boot
Data Warehousing and Platforms: Snowflake, Redshift, PostgreSQL, TimescaleDB, Delta Lake, data modeling, dimensional modeling, schema design, data architecture
Data Pipelines and ETL/ELT: PySpark, Databricks, Apache Kafka, Apache Airflow, dbt, streaming and batch ingestion, multi format data structuring (Excel, PDF, raw sources)
Data Quality and Governance: Great Expectations, data lineage, schema drift detection, data integrity frameworks, audit ready reporting, CI/CD test gates
Cloud Platforms: AWS (Lambda, S3, EC2, RDS, Kinesis), Azure, GCP, Databricks on cloud, cloud data platform migration
MLOps and Observability: Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Prometheus, Grafana, drift detection, anomaly detection, Isolation Forest
ML and Modeling: Classification, regression, feature engineering, EDA, model evaluation, XGBoost, A/B testing, MLflow